

BUILD FOR INDIA - SIH Internal Hackathon 2025

Theme: Miscellaneous

Team Name: DreamBit

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Problem Statement:

Communication is all about connecting but for those who use Indian Sign Language (ISL) spoken words can create gaps. Public announcements can become confusing classrooms can feel far away and conversations can seem one-sided because most people who don't know ISL cannot understand it and there is no common link between spoken words and signs. This barrier doesn't just stop words; it limits chances access and a sense of belonging leaving deaf and hard-of-hearing individuals feeling alone in education healthcare jobs and everyday life.

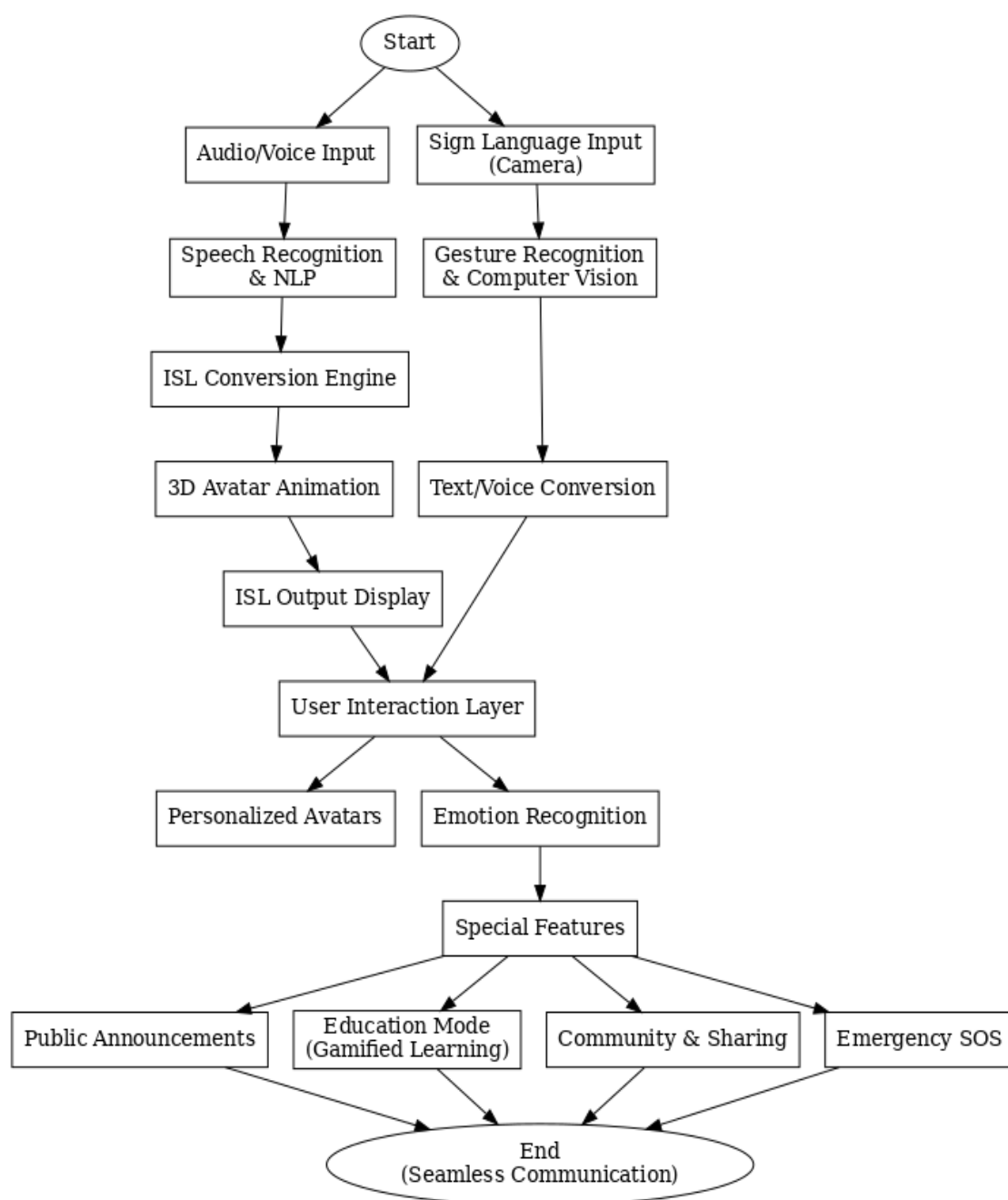
The goal is to create a real-time translator that connects spoken language with ISL breaking down these barriers so that every voice can be heard and every sign can be understood.

Proposed Solution:

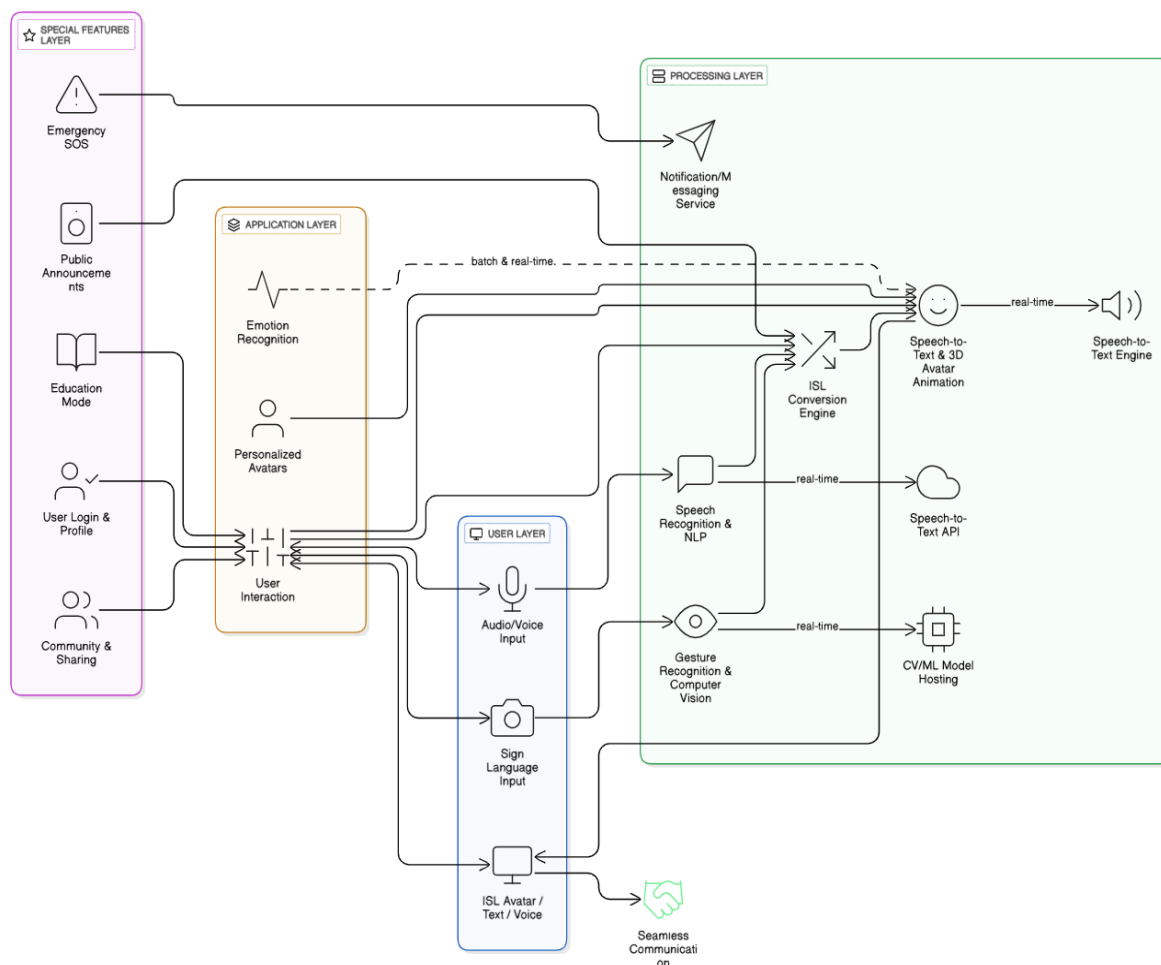
Breaking Barriers with AI is an interactive web application aimed at transforming communication by converting English and Hindi audio-visual content into vibrant Indian Sign Language (ISL) animations—and vice versa. Picture listening to a lecture, film, or dialogue that springs to life immediately in ISL, or seeing sign gestures easily transcribe into speech and text for instantaneous two-way communication. Driven by Real-Time Audio → ISL Conversion on 3D animated avatars, Sign-to-Text/Voice Recognition by camera capture, and Context-Aware Emotion Recognition for precise tone delivery, the platform is an actual communication hub for inclusivity. For public use, in addition to personal use, it provides Public Spaces Plug-ins for railways, airports, and hospitals, immediately translating announcements into ISL on display, and Personalized ISL Avatars to establish emotional rapport. Augmented with an Education Mode (gamified learning of ISL), Community Features for assistance and social interaction, and a one-tap SOS Button with live location tracking, it guarantees both empowerment and security. With intuitive, easy-to-use

UI/UX, regional ISL adaptability, and uses in the realms of education, public services, workplaces, and society, this innovation is more than a tool but a movement towards "Communication for All"—breaking the silence barrier and promoting real inclusivity.

Flowchart:



System Architecture:



Tech Stack:

- **Frontend (User Experience)**
 - React Native + React.js → Cross-platform app + web.
 - Three.js / Babylon.js → Interactive 3D ISL avatars.
 - ARCore / ARKit → AR-based ISL learning (avatars in classrooms/public spaces).
 - Accessibility-first → Haptic feedback, large icons, color-blind friendly UI.
- **Backend & Cloud:**
 - FastAPI + Node.js → Real-time APIs & streaming.
 - PostgreSQL + Redis → Secure + fast DB & caching.
 - Firebase → Authentication, push notifications, real-time sync.
 - GraphQL → Flexible API for mobile/web queries.
- **AI/ML Intelligence:**
 - Whisper / Vosk → Speech-to-Text (online & offline).
 - MediaPipe + TensorFlow Lite → Real-time sign gesture recognition.
 - NLP (spaCy / HuggingFace) → Map text ↔ ISL dictionary.
 - Emotion AI → Detect tone/mood → avatar signs with expressions.

Research / Early Work:

In our initial work we have examined and found effective technologies to create a solid base for the system. For sign recognition (Sign-to-Text) we are using Google's MediaPipe a dependable framework for tracking hands and bodies in real time which ensures accurate and efficient recognition. For converting audio to sign language we use browser-based speech-to-text APIs to capture spoken words and match keywords with pre-recorded sign language videos allowing smooth two-way communication. Our method for gamified learning takes inspiration from successful programs like Duolingo using quizzes and interactive feedback to make learning fun and effective. Additionally for personalized avatars and emergency support features we plan to integrate third-party solutions like Ready Player Me for customizing avatars as well as standard device geolocation and SMS APIs for emergency help. These choices show our research-based approach using strong existing technologies to speed up development while ensuring reliability.

Implementation Plan :

1- Requirement Analysis

- Objective: Create a real-time translator for spoken language and Indian Sign Language (ISL).
- Scope: Improve access in education, healthcare, workplaces, and public communication.
- Resources: Use cloud servers, APIs (like MediaPipe and Speech-to-Text), a video dataset, development tools, and a testing environment.

2- Team Formation & Role Allocation

- Technical Team: Focus on speech-to-text, gesture recognition, and avatar integration.
- Design Team: Work on user interface/user experience (UI/UX), flowcharts, and diagrams.
- Research & Content Team: Curate datasets and create keyword-sign mappings.
- Testing & Deployment Team: Handle debugging and cloud deployment.
- Communication & Documentation Team: Prepare reports, presentations, and manage coordination.
- Hold weekly sprint meetings to monitor progress.

3- Resource Planning

- Development Tools: Use IDEs, GitHub, and Figma.
- Frameworks/APIs: Implement Google MediaPipe, Browser STT, and Ready Player Me.
- Infrastructure: Ensure high-speed internet, webcams, and microphones are available.
- Collaboration Tools: Use Slack or Discord, Trello, and Google Meet for team communication.

4- Problem Statement Finalization

- Challenge: Develop a real-time translator that connects spoken language with ISL.
- Criteria: Ensure the solution is practical, relevant, and impactful in real-world situations.

5- Communication & Guidelines

- Standardize coding practices and how datasets are used.
- Utilize version control with GitHub.
- Create structured templates for reporting and documentation.
- Provide reference materials, including sample diagrams and system architecture.

6- Execution Meetings

- Conduct bi-weekly review meetings to update on module progress.
- Monitor advancements in gesture recognition, speech mapping, and avatar development.
- Encourage open discussions to quickly resolve any issues.
- Set midway checkpoints to ensure deadlines are met.

7- Event Day Preparation

- Prepare for a live demonstration of sign-to-text and audio-to-sign translation.
- Walk through the architecture using flowcharts and system design.
- Hold a mentorship session for feedback before the final pitch.
- Develop a backup plan that includes pre-recorded demos, failover servers, and technical support.

Reusability & Documentation:

The system is designed using a modular service-oriented architecture. Key modules such as speech-to-text, the ISL dictionary, gesture recognition, and SOS are available through REST/GraphQL APIs. This allows them to be integrated into other applications like e-learning platforms, public announcement systems, or healthcare tools.

Machine learning models for sign recognition and emotion detection are packaged with TensorFlow Lite/ONNX, making it easy to use across different devices. The ISL dictionary is stored in a flexible JSON/CSV format, which makes it simple to add new regional or international sign languages.

The documentation includes automatically generated API references (Swagger/Postman), architecture diagrams, and setup scripts using Docker to ensure consistency. The project is hosted on GitHub, featuring a clear README, guidelines for contributing, and an open-source license to encourage community-driven improvements.

References:

1. M. J. C. Samonte, R. A. Gazmin, J. D. S. Soriano and M. N. O. Valencia, "BridgeApp: An Assistive Mobile Communication Application for the Deaf and Mute," 2019 International Conference on Information and Communication Technology Convergence (ICTC), Jeju, Korea (South), 2019, pp. 1310-1315, doi: 10.1109/ICTC46691.2019.8939866. <https://ieeexplore.ieee.org/document/8939866>
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3.[https://www.researchgate.net/publication/374147369 Deaf students and the challenges they face in higher education](https://www.researchgate.net/publication/374147369_Deaf_students_and_the_challenges_they_face_in_higher_education)

4.[https://www.researchgate.net/publication/346575410 The learning social and economic challenges facing the deaf and hearing-impaired individuals](https://www.researchgate.net/publication/346575410_The_learning_social_and_economic_challenges_facing_the_deaf_and_hearing-impaired_individuals)

5.islrtc.nic.in